

Firms, Hiring, Management, And Organizational Response To New Technology

AI Adoption As A Firm-Side Labor Problem

Special Topic 5: Technology, Innovation, and Labor – Week 4

Central question

- When a firm adopts a new technology, what changes inside the firm?
- Which jobs are redesigned, which workers are trained, and which vacancies are posted?
- How do supervision, monitoring, delegation, promotion, and internal mobility change?
- Week 4 discipline: adoption is not complete until work is reorganized.

Why this is labor economics

- Technology is implemented through vacancies, task assignments, managers, teams, and contracts.
- The same AI tool can substitute for some tasks and augment others.
- Firm choices determine whether incumbents are retrained, replaced, monitored, or promoted.
- Worker and manager attitudes can shape usage, resistance, trust, and measured productivity.

From adoption to reorganization

Margin	What changes	Labor object
Acquisition	buy, build, license, or hire for AI	investment and adoption
Implementation	connect data, workflows, incentives	management and training
Usage	workers or managers rely on the tool	task performance and quality
Reorganization	jobs, teams, authority, ladders change	labor demand and incidence

- Exposure, adoption, usage, and redesign are different empirical treatments.

Adoption as an organizational choice

$$Adopt_{ft} = 1 \left[E_t \left(\Pi_{ft}^{AI}(K_{ft}^O, H_{ft}, M_{ft}, A_{ft}) \right) - \Pi_{ft}^0 - C_{ft}^{adopt} - C_{ft}^{org} > 0 \right]$$

- K^O : organizational capital; H : workforce skills and hierarchy.
- M : management practice; A : worker and manager acceptance or resistance.
- Adoption depends on complementary investments, not only the tool price.

Adopters are selected

- Early AI adopters may already be larger, more productive, more skilled, and better managed.
- Adoption estimates mix the technology with firm capability, timing, and workforce composition.
- A vacancy or resume measure captures intended skill demand, not necessarily realized usage.
- A survey adoption measure may capture aspiration, experimentation, or full workflow redesign.

AI as three labor objects

Margin	Mechanism	Labor implication
Substitution	AI performs tasks cheaply	task displacement or role compression
Augmentation	AI raises worker productivity	higher output, quality, or scale
Delegation	AI changes who decides	authority, monitoring, and evaluation shift

- The same tool can substitute, augment, and reallocate control at the same firm.

Firm labor-demand decomposition

$$\Delta L_f = \Delta L_f^{sub} + \Delta L_f^{aug} + \Delta L_f^{scale} + \Delta L_f^{org}$$

- Substitution: fewer workers needed for automated tasks.
- Augmentation: worker-performed tasks become more productive.
- Scale: lower costs or higher quality raise output demand.
- Organization: hierarchy, monitoring, teams, and job boundaries change.

Job redesign and task reassignment

- Firms rebundle tasks into redesigned jobs rather than replacing occupations one for one.
- Workers may shift toward validation, exception handling, customer communication, or system maintenance.
- The key question is whether old firm-specific experience remains valuable after redesign.
- Measured wage gains can reflect recomposition toward new workers rather than incumbent gains.

Hiring, screening, and internal labor markets

- Hiring response: AI engineers, data roles, compliance roles, domain experts, and AI-fluent generalists.
- Screening response: more emphasis on autonomy, judgment, digital fluency, and auditing machine output.
- Internal response: training, task reassignment, lateral moves, promotion redesign, or exit.
- Incumbent training and hiring around incumbents are different labor-market mechanisms.

Principal-agent problems

Margin	Firm or manager objective	Worker concern
Monitoring	observe effort and output	privacy, autonomy, gaming
Evaluation	standardize performance metrics	fairness and opacity
Delegation	speed and consistency	loss of discretion
Replacement risk	lower costs and reassign tasks	job security and de-skilling

- AI changes not only productivity, but who controls information and who bears risk.

AI attitudes and algorithm aversion

- Workers may resist AI because of replacement risk, monitoring, unfair evaluation, or loss of control.
- Managers may resist delegation because of overconfidence, liability, discretion, or status.
- Attitudes affect measured adoption: access, self-reported use, mandated use, and trusted use differ.
- Algorithm aversion is a labor issue when it changes hiring, screening, promotion, or supervision.

Design	Identifies well	Main caution
Worker survey + outcomes	trust, fear, usage, retention	attitudes are endogenous
Management survey + admin data	beliefs and implementation strategy	manager quality confounds
Randomized rollout	access or encouragement effects	short-run horizon
Vignette or conjoint	fairness, transparency, control	stated preference
Usage logs + surveys	realized use and acceptance	compliance vs belief
Employer-worker panel	worker incidence around adoption	adopter selection

What a strong design separates

- ① Technology availability or exposure.
- ② Firm adoption and implementation timing.
- ③ Actual worker or manager usage.
- ④ Attitudes: trust, fairness, replacement concern, delegation willingness.
- ⑤ Organizational redesign: hiring, training, monitoring, promotion, and job boundaries.
- ⑥ Worker outcomes: productivity, wages, separations, autonomy, job quality, and mobility.

- Reproduce: build a synthetic firm AI-adoption panel with workforce composition and hierarchy.
- Diagnose: separate selection, acquisition, usage, substitution, augmentation, scale, and redesign.
- Transfer: add worker and manager attitudes inspired by algorithm-aversion designs.
- Output: compact tables that distinguish adoption, usage, sentiment, productivity, and worker incidence.
- Goal: practice the design logic without confidential firm records or official replication data.

- Week 4 studies how firms implement technology inside jobs and internal labor markets.
- Week 5 asks how firm-level adoption aggregates into inequality, rents, market power, and institutions.
- The bridge is distributional: firm capabilities and worker access shape who captures technology gains.

- Firm adoption is an organizational labor problem, not a simple capital purchase.
- AI can substitute, augment, and reallocate authority at the same time.
- Hiring, training, monitoring, promotion, and job redesign determine worker incidence.
- Worker and manager attitudes are adoption frictions and inequality channels.
- Strong empirical work separates adoption, usage, sentiment, redesign, and outcomes.